RAJALAKSHMI ENGINEERING COLLEGE

Dhivya Sri G 251001055-2025-IT

D2

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|---|---|
| **Started on** | Friday, 9 October 2026, 7:22 PM |
| **State** | Finished |
| **Completed on** | Friday, 9 October 2026, 7:23 PM |
| **Time taken** | 1 min 13 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| s1 | | a | g | **g** | **t** | **a** | **b** |
| s2 | **g** | x | **t** | **x** | **a** | **y** | **b** |

**The length is 4**

Solveing it using Dynamic Programming

**For example:**

| Input | Result |
|---|---|
| aab<br>azb | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

int main() {
    char a[1000], b[1000];
    int dp[1000][1000];
    int i, j, m, n;

    scanf("%s %s", a, b);

    m = strlen(a);
    n = strlen(b);

    for (i = 0; i <= m; i++) {
        for (j = 0; j <= n; j++) {
            if (i == 0 || j == 0)
                dp[i][j] = 0;
            else if (a[i - 1] == b[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;
            else if (dp[i - 1][j] > dp[i][j - 1])
                dp[i][j] = dp[i - 1][j];
            else
                dp[i][j] = dp[i][j - 1];
        }
    }

    printf("%d\n", dp[m][n]);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | aab<br>azb | 2 | 2 | ✔ |
| ✔ | ABCD<br>ABCD | 4 | 4 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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